



Type 2 Diabetes Mellitus in Children and Adolescents

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Type 2 diabetes mellitus is characterized by insulin resistance and inadequate insulin secretion relative to physiologic needs. Early symptoms are related to hyperglycemia and include polydipsia, polyphagia, polyuria, and blurred vision. Diagnosis is by measuring plasma glucose. Treatment is diet, exercise, and medications that reduce glucose levels, including metformin, insulin, and non-insulin injectable and oral medications. Later complications include vascular disease, peripheral neuropathy, nephropathy, and predisposition to infection. Complications can be delayed or prevented with adequate glycemic control.

Rising prevalence of childhood obesity has led to an increase in the incidence of childhood type 2 diabetes, and in increase in the proportion of type 2 relative to type 1 diabetes diagnoses in children ([1](#), [2](#)). Type 2 diabetes is typically diagnosed in early adolescence with onset of puberty, when insulin resistance naturally increases and there is rapid growth and pubertal changes. (See also [obesity in children](#) and [obesity in adolescents](#).)

Approximately 75% of children with type 2 diabetes have obesity ([3](#)). However, there is considerable heterogeneity, and the relationship between obesity and age at onset of type 2 diabetes is less clear in some ethnicities (eg, Asian, particularly Japanese, children) ([3](#)).

General references

1. [Gesuita R, Eckert AJ, Besançon S, et al](#). Frequency and clinical characteristics of children and young people with type 2 diabetes at diagnosis from five world regions between 2012 and 2021: data from the SWEET Registry. *Diabetologia*. 2025;68(1):82-93. doi:10.1007/s00125-024-06283-5
2. [Wagenknecht LE, Lawrence JM, Isom S, et al](#). Trends in incidence of youth-onset type 1 and type 2 diabetes in the USA, 2002-18: results from the population-based SEARCH for Diabetes in Youth study. *Lancet Diabetes Endocrinol*. 2023;11(4):242-250. doi:10.1016/S2213-8587(23)00025-6
3. [Cioana M, Deng J, Nadarajah A, et al](#). The Prevalence of Obesity Among Children With Type 2 Diabetes: A Systematic Review and Meta-analysis. *JAMA Netw Open*. 2022;5(12):e2247186.

Etiology of Type 2 diabetes in Children and Adolescents

In type 2 diabetes, the pancreas produces insulin, but there are varying degrees of insulin resistance, and insulin secretion is inadequate to meet the increased demand caused by insulin resistance (*relative insulin deficiency*).

Onset of type 2 diabetes often coincides with the peak of physiologic pubertal insulin resistance, which may lead to symptoms of hyperglycemia in previously compensated adolescents.

The cause of type 2 diabetes is not autoimmune destruction of beta-cells but rather a complex interaction between many genes and environmental factors, which differ among different populations and patients.

Type 2 diabetes in children is different than type 2 diabetes in adults ([1](#)). In children, decline in beta-cell function and development of diabetes-related complications are accelerated.

Risk factors for type 2 diabetes include ([2](#), [3](#), [4](#)):

- Obesity (less common in children of Asian descent)
- Native American, Canadian First Nation, Indigenous Australian, Black, Hispanic, East and South Asian, Middle Eastern, and Pacific Islander heritage
- Those with [hypertension](#), [hyperlipidemia](#), [obstructive sleep apnea](#), acanthosis nigricans, [hepatic steatosis](#), [polycystic ovary syndrome](#), or were [small-for-gestational-age at birth](#)
- Male sex
- First-degree relative with type 2 diabetes
- Maternal diabetes (including gestational diabetes) or obesity during pregnancy
- Excessive intake of sugar-sweetened beverages
- Low physical activity

Etiology references

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2. [Bellou V, Belbasis L, Tzoulaki I, Evangelou E](#). Risk factors for type 2 diabetes mellitus: An exposure-wide umbrella review of meta-analyses. *PLoS One*. 2018;13(3):e0194127. doi:10.1371/journal.pone.0194127
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Pathophysiology of Type 2 Diabetes in Children and Adolescents

In type 2 diabetes, there is usually enough insulin function to prevent [diabetic ketoacidosis](#) (DKA) at diagnosis, but children with type 2 diabetes can sometimes present with DKA (approximately 5 to 6%) ([1](#), [2](#)) or, less commonly, [hyperglycemic hyperosmolar state](#) (HHS), also referred to as hyperosmolar hyperglycemic nonketotic syndrome (HHNK), in which severe hyperosmolar dehydration occurs. HHS most often occurs during a period of stress or infection, with nonadherence to treatment regimens, or when glucose metabolism is further impaired by medications (eg, glucocorticoids). Other metabolic derangements associated with insulin resistance can be present at diagnosis of type 2 diabetes and include:

- Dyslipidemia (leading to [atherosclerotic cardiovascular disease](#))
- [Hypertension](#)
- [Polycystic ovary syndrome](#)
- [Obstructive sleep apnea](#)
- [Metabolic dysfunction-associated steatohepatitis](#) (fatty liver)

Atherosclerosis in patients with type 2 diabetes begins in childhood or adolescence and markedly increases risk of cardiovascular disease.

Pathophysiology references

1. [Dabelea D, Rewers A, Stafford JM, et al.](#) Trends in the prevalence of ketoacidosis at diabetes diagnosis: the SEARCH for diabetes in youth study. *Pediatrics*. 2014;133(4):e938-e945. doi:10.1542/peds.2013-2795
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Symptoms and Signs of Type 2 Diabetes in Children and Adolescents

In type 2 diabetes, the clinical presentation varies widely. Children are often asymptomatic or minimally symptomatic, and their condition may be detected only on routine testing. However, some children present with symptomatic hyperglycemia (polyuria and urinary frequency, polydipsia), [hyperglycemic hyperosmolar state](#), or [DKA](#).

Even without symptoms of hyperglycemia, individuals with type 2 diabetes frequently have overweight or obesity, as well as physical findings suggestive of insulin resistance (eg, acanthosis nigricans, skin tags).

Dermatologic Signs of Insulin Resistance



Acanthosis Nigricans

Acanthosis nigricans is skin thickening and pigmentation most typically developing in the axilla and nape of the neck (top); in people with darkly pigmented skin, the skin may have a leathery appearance (bottom). It is most often a skin manifestation of impaired glucose tolerance.

Photos provided by Thomas Habif, MD.



Acanthosis Nigricans (Elbow Fold)

This photo shows acanthosis nigricans (hyperpigmented skin) in the elbow fold due to insulin resistance.

RICHARD USATINE MD / SCIENCE PHOTO LIBRARY



Acanthosis Nigricans (Skin Tags)

This photo shows acanthosis nigricans (hyperpigmented skin) and multiple skin tags due to insulin resistance.

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Pearls & Pitfalls

- Despite the common misconception, diabetic ketoacidosis can occur in children with type 2 diabetes.

Diagnosis of Type 2 Diabetes in Children and Adolescents

- Measurement of abnormal glucose or glucose tolerance meeting diagnostic criteria (see table [Diagnostic Criteria for Diabetes Mellitus and Prediabetes](#))
- Determination of diabetes type (eg, type 1, type 2, monogenic)
- C-peptide, insulin, and pancreatic autoantibody testing
- Additional laboratory tests to screen for metabolic comorbidities (eg, liver function tests, fasting lipid profile, spot urine microalbumin to creatinine ratio)

Diagnosis of diabetes and prediabetes is based on abnormal glucose measurements or glucose tolerance meeting the diagnostic criteria used by the American Diabetes Association (see [Diagnosis of Diabetes in Children](#)) (1).

Children without autoantibodies, who are older than 10 years (or have started puberty) at diagnosis, are obese, or have evidence of metabolic co-morbidities are generally treated as having type 2 diabetes (2).

Prediabetes is diagnosed when measures of glucose are between the normal range and the range diagnostic for diabetes (see table [Diagnostic Criteria for Diabetes Mellitus and Prediabetes](#)). It is sometimes considered a precursor or transitional state prior to the development of type 2 diabetes, although data about the rate of progression or risks specific to prediabetes are limited in children (2).

Screening for type 2 diabetes

Screening is recommended beginning at age 10 or the onset of puberty in children who have overweight and have ≥ 1 additional [risk factor](#) or conditions associated with insulin resistance. Conditions associated with insulin resistance include:

- Acanthosis nigricans
- [Hypertension](#)
- [Dyslipidemia](#)
- [Polycystic ovary syndrome](#)
- [Small for gestational age](#)

- [Large for gestational age](#)
- Use of atypical antipsychotic agents with rapid weight gain

Screening may consist of a fasting glucose, a glycosylated hemoglobin (HbA1C), and/or an oral glucose tolerance test. Repeat screening should be done every 2 to 3 years, or more frequently if the overall cardiometabolic risk profile is worsening.

Screening for comorbidities and complications

Monitoring for **acute complications** of both hypoglycemia and hyperglycemia (including DKA and HHS) is discussed in [Glycemic Control](#); their management is discussed in [Complications of Diabetes in Children and Adolescents](#). For more detail, see [Type 2 Diabetes Mellitus - Complications](#) and [Long-Term Complications of Diabetes Mellitus](#).

As with adults with type 2 diabetes (and unlike children with type 1 diabetes), screening for **long-term (vascular) complications** in patients with type 2 diabetes begins at diagnosis. Screening includes evaluation of feet, eyes, kidney function and proteinuria, and blood pressure.

Patients with type 2 diabetes (unlike those with type 1 diabetes, in whom [complications](#) develop over many years) often have **comorbidities** at diagnosis, such as metabolic dysfunction-associated fatty liver disease, hyperlipidemia, and hypertension. Patients should have liver tests, fasting lipid profile, metabolic profile with measures of glomerular filtration rate, and a spot urine protein-to-creatinine ratio performed at the time of diagnosis ([2](#)). Blood pressure measurement and mental health screening should be performed at diagnosis and at every diabetes-related visit. Eye and foot examinations should be performed at diagnosis and annually. Children with clinical findings suggestive of complications should also be tested:

- Obesity: Test for [metabolic dysfunction-associated steatohepatitis](#)
- Daytime somnolence or snoring while sleeping: Test for [obstructive sleep apnea](#)
- Hirsutism, acne, or menstrual irregularities: Test for [polycystic ovary syndrome](#)

Diagnosis references

1. [American Diabetes Association Professional Practice Committee](#). 2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S27-S49. doi:10.2337/dc25-S002
2. [Shah AS, Barrientos-Pérez M, Chang N, et al](#). ISPAD Clinical Practice Consensus Guidelines 2024: Type 2 Diabetes in Children and Adolescents. *Horm Res Paediatr*. 2024;97(6):555-583. doi:10.1159/000543033

Treatment of Type 2 Diabetes in Children and Adolescents

-
- Diet
 - Exercise

- Antihyperglycemic medication, sometimes including insulin
- Glycemic control

As with type 1 diabetes, glycemic control is the cornerstone of treatment. This may be accomplished with some combination of [lifestyle modifications](#), including dietary modification, physical activity, and weight management, oral antihyperglycemic medication, and insulin. In addition, there is an added emphasis on the detection and management of associated cardiometabolic disorders and chronic vascular complications.

Children with type 2 diabetes require the same attention to diet and weight control and recognition and management of dyslipidemia and hypertension as do adults, although screening recommendations are not identical ([1](#)). Most children with type 2 diabetes have obesity ([2](#)), so lifestyle modification is the cornerstone of therapy. Pharmacotherapy may also be indicated.

Diet and exercise

All children with type 2 diabetes should receive comprehensive diabetes self-management education and support, including nutrition education, preferably with the involvement of a dietitian experienced in the management of children with diabetes ([1](#), [3](#)). Most patients should be encouraged to lose weight and thus increase insulin sensitivity.

Steps to improve diet, manage caloric intake, and improve physical activity include:

- Eliminate sugar-containing drinks and foods made of refined, simple sugars (eg, processed candies, high fructose corn syrups).
- Discourage skipping meals and encourage eating meals on a schedule (preferably as a family, if possible, and without distractions from other activities, eg, television, computer, or video games).
- Avoid grazing on food throughout the day.
- Control portion size.
- Limit high-fat, high-calorie foods in the home.
- Increase fiber intake by eating more fruits and vegetables.
- Increase physical activity to 60 minutes of moderate to vigorous physical activity at least 3 days per week (preferably 5 to 7 days per week).
- Limit screen time to < 2 hours a day, including television, noneducational computer time, cell phones and other handheld devices, and video games.

For children age 3 to 13 years, the formula 1000 calories + (100 × child's age in years) provides an estimate of daily caloric need similar to published guidelines ([4](#)).

Metabolic surgery (also called bariatric or obesity surgery) may be considered for weight loss in children 12 years and older with type 2 diabetes and a body mass index ≥ 35 kg/m² ([3](#)).

Pharmacologic treatment

For asymptomatic patients with mild hyperglycemia, lifestyle therapy can be tried, but if glycemic targets are not met within approximately 3 to 6 months, metformin should be considered.

Metformin

Metformin is an insulin sensitizer and is the most common first-line oral antihyperglycemic medication given to patients < 18 years of age. Metformin is started as monotherapy when the initial HbA1C level is < 8.5% (< 69 mmol/mol) without acidosis or ketosis, and is used in conjunction with nonpharmacologic therapy including dietary modification and exercise (3).

Metformin should be started at a low dose and taken with food to prevent nausea and abdominal pain. The dose is increased stepwise to the maximal target dose (2 g/day) over a period of 3 to 6 weeks. If available, extended-release forms of metformin may lessen gastrointestinal adverse effects in some patients who do not tolerate standard formulations of this medication.

The goal of treatment is an HbA1C level at least < 7% (< 53 mmol/mol) and preferably < 6.5% (< 48 mmol/mol). If this cannot be achieved with metformin alone, basal insulin or a glucagon-like peptide 1 (GLP-1) receptor agonist should be started. Unfortunately, in approximately half of adolescents with type 2 diabetes metformin monotherapy ultimately fails and insulin is required.

Insulin

Insulin is also initiated in children who present with more severe type 2 diabetes (HbA1C > 8.5% [> 69 mmol/mol]) (3); glargine, detemir, or premixed insulin can be used.

Children with DKA or HHS usually require intravenous insulin initially and then are transitioned to subcutaneous insulin (3). Once acidosis resolves, metformin can be initiated.

Insulin requirements may decline rapidly during the initial weeks of treatment as endogenous insulin secretion increases; insulin often can be discontinued several weeks after regaining acceptable metabolic control.

However, a patient receiving combination therapy (sometimes including basal insulin) who is not achieving glycemic targets may also require prandial insulin boluses (3).

Other antihyperglycemic medications in children

If patients do not meet targets using dual therapy with metformin and basal insulin, **GLP-1 receptor agonists** may be added as part of intensification therapy. Oral medications also may be considered.

Liraglutide, extended-release exenatide, and dulaglutide are GLP-1 receptor agonists that can be used in children > 10 years of age with type 2 diabetes and can help reduce HbA1C levels (3). Injectable semaglutide is another GLP-1 receptor agonist used for treatment of obesity in patients > 12 years of age (5). These injectable noninsulin antihyperglycemic medications enhance glucose-dependent insulin secretion and slow gastric emptying.

Liraglutide is given as a daily injection, whereas extended-release exenatide, dulaglutide, and semaglutide are given as weekly subcutaneous injections, which may improve patient adherence. All of

these medications promote weight loss, likely through the effect of delayed gastric emptying and appetite reduction. They are titrated to treatment doses over a period of weeks to minimize the common gastrointestinal adverse effects, especially nausea and vomiting. GLP-1 agonists can be used if metformin is not tolerated or added on if HbA1C target levels are not achieved with metformin alone within 3 months. GLP-1 agonists may be used before the initiation of insulin to promote weight loss as well as glycemic control.

Empagliflozin, canagliflozin, and dapagliflozin are **sodium-glucose cotransporter-2 (SGLT2) inhibitors** used for children > 10 years of age with type 2 diabetes ([1](#), [3](#), [6](#)). SGLT2 is a glucose transporter found in the proximal tubule of the kidneys. It is responsible for approximately 90% of filtered glucose reabsorption. SGLT2 inhibitors work by blocking the coupled reabsorption of sodium and glucose from the proximal tubules, leading to increased renal excretion of glucose and lower blood glucose levels in patients with type 2 diabetes. These medications are contraindicated in patients with end-stage renal disease or who are on dialysis. They can increase the risk of DKA, in some cases with normal blood glucose levels because of increased renal excretion of glucose. Adverse effects of these medications include increased incidence of urinary tract and genital yeast infections.

Glycemic control and HbA1C target levels

Patients with type 2 diabetes usually self-monitor blood glucose levels less frequently than patients with type 1 diabetes, but frequency varies depending on the type of therapy used, fasting and postprandial glucose levels, degree of glycemic control deemed achievable, and the available resources.

Children and adolescents taking multiple daily insulin injections, those who are ill, and those with suboptimal control should monitor glucose levels at least 3 times a day ([7](#)). Those who are on stable regimens of metformin plus long-acting insulin only (ie, no short-acting bolus insulin) who are meeting their targets without hypoglycemia, can monitor less frequently, typically 2 times a day (fasting and 2 hours postprandial). The frequency of monitoring should increase if glycemic control targets are not being met, during illness, or when symptoms of hypoglycemia or hyperglycemia develop. Children and adolescents with type 2 diabetes on insulin regimens with multiple daily injections or insulin pumps should be offered CGM systems similar to systems used by those with type 1 diabetes ([1](#)).

The recommended **HbA1C target level for type 2 diabetes** in most children and adolescents is < 6.5% (< 48 mmol/mol) ([1](#), [3](#)). As with all glycemic targets, levels are individualized based on the patient's metabolic requirements and characteristics. For those at higher risk of hypoglycemia a higher target (< 7% [< 53 mmol/mol]) may be appropriate.

HbA1C levels should be measured every 3 months in most children with type 2 diabetes, especially if insulin is being used or metabolic control is suboptimal. In some children with stable glucose levels, levels can be measured twice a year.

Children with type 2 diabetes who do not meet HbA1C and/or fasting glucose targets are candidates for intensified therapy (eg, with insulin, including bolus insulin dosing, [GLP-1 receptor agonists](#), and/or SGLT-2 inhibitors).

Treatment references

1. [American Diabetes Association Professional Practice Committee](#). 14. Children and Adolescents: Standards of Care in Diabetes-2025. *Diabetes Care*. 2025;48(1 Suppl 1):S283-S305. doi:10.2337/dc25-S014
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7. [Copeland KC, Silverstein J, Moore KR, et al](#): Management of newly diagnosed type 2 Diabetes Mellitus (T2DM) in children and adolescents. *Pediatrics* 131(2):364-382, 2013. doi: 10.1542/peds.2012-3494

Key Points

- Type 2 diabetes is caused by insulin resistance and relative insulin deficiency due to a complex interaction among many genetic and environmental factors (particularly obesity); it is increasing in frequency in children and is usually diagnosed after puberty.
- Most children have symptomatic hyperglycemia without acidosis, with urinary frequency, polydipsia, and polyuria; rarely children with type 2 diabetes may present with diabetic ketoacidosis or hyperglycemic hyperosmolar state.
- Screen at-risk children for type 2 diabetes or prediabetes.
- Children with type 2 diabetes are initially treated with metformin and/or insulin; most children requiring insulin at diagnosis can be successfully transitioned to metformin monotherapy, but about half eventually require insulin treatment.
- GLP-1 agonists or SGLT-2 inhibitors can be used in combination with metformin to improve glycemic control.
- Children are at risk of microvascular and macrovascular complications of diabetes, which must be evaluated by regular screening tests.

